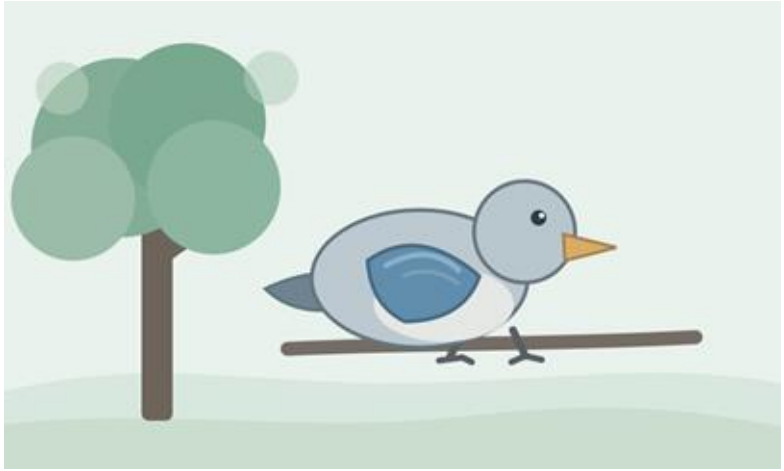
Concept Bottleneck Models

- Motivation
- VLM guidance
- Procedure Video CBM
- Research directions



Jimmy
10.22

Why a Linear Model Is Interpretable

$$y = w1*x1 + w2*x2 + w3*x3 + \ldots + wn*xn + b$$

w1*x1	w2*x2	w3*x3	...
Feature 1 contribution	Feature 2 contribution	Feature 3 contribution	

1. Sign of w tells direction

positive weight -> increases output
negative weight -> decreases output

2. Magnitude of w tells strength

larger absolute weight -> stronger effect
smaller absolute weight -> weaker effect

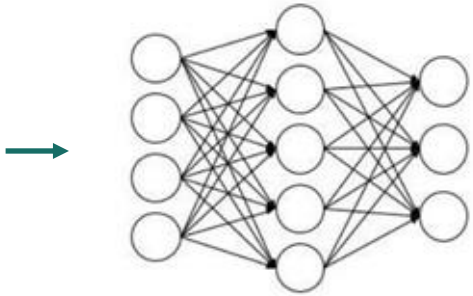
3. Each term is directly inspectable

the prediction can be decomposed
into per-feature contributions

A black-box model is uninterpretable

Bird classification





predicts Bird B

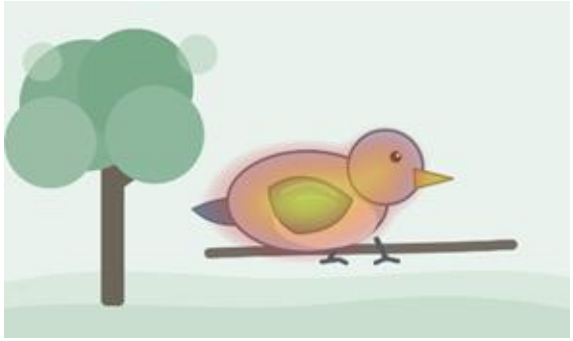
Grad-CAM: where is the evidence?

Gradient-based attribution highlights image regions supporting a class score.

INPUT IMAGE



ILLUSTRATIVE HEATMAP



Model output

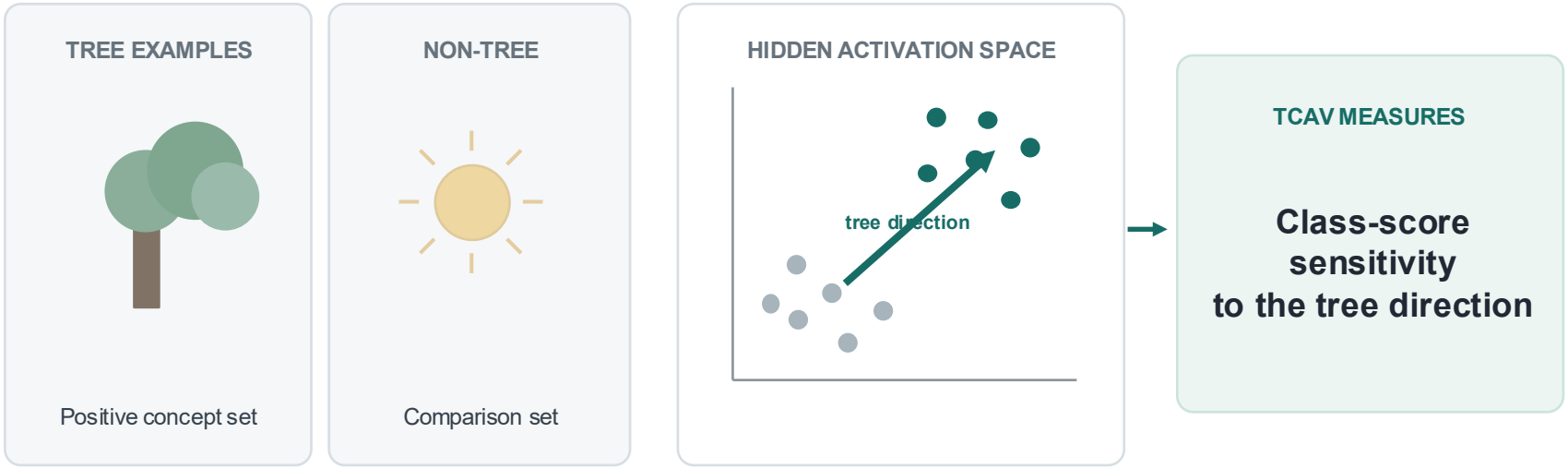
Bird B

The explanation describes a trained model; it does not change the prediction.



TCAV: which human concept matters?

A Concept Activation Vector (CAV) represents a specified concept in activation space.



- More semantic than a heatmap — but still an analysis of a trained model.

CBM makes concepts part of the forward pass

In a strict CBM, the final classifier receives concept predictions instead of image features.

STANDARD MODEL

Image features → Class predictor → Bird A

CBM



→ Concept predictor f → Predicted concepts

red wing = Yes ✗
short beak = Yes
dark head = No

→ Label predictor g → Bird A

$$\hat{c} = f(x) \qquad \hat{y} = g(\hat{c})$$

Crucial: there is no separate image → class bypass in a strict CBM.

Koh et al., ICML 2020

Concept intervention: explain, then correct

At test time, a user can override an incorrect concept and rerun the label predictor.

BEFORE CORRECTION

Predicted concept vector

red wing = Yes

short beak = Yes

Bird A

AFTER CORRECTION

Corrected concept vector

red wing = No

short beak = Yes

Bird B

The classifier g acts on the revised concepts; the whole model is not retrained.

Pseudo Concept Supervision for CBMs

TRAINING TIME

INFERENCE TIME

1

Raw data



No human concept labels.

2

Teacher model

pretrained VLM / LLM



Caption images

Extract attributes

Detect objects

3

Pseudo concepts

red wing

long beak

tree bg

These pseudo labels supervise the CBM concept layer.

4

CBM concept predictor

(student)



Light weight models like MLP learns to predict concept presence from input

5

Concept bottleneck → task prediction

concepts

c1 red wing

c2 long beak

c3 tree bg

...

task head

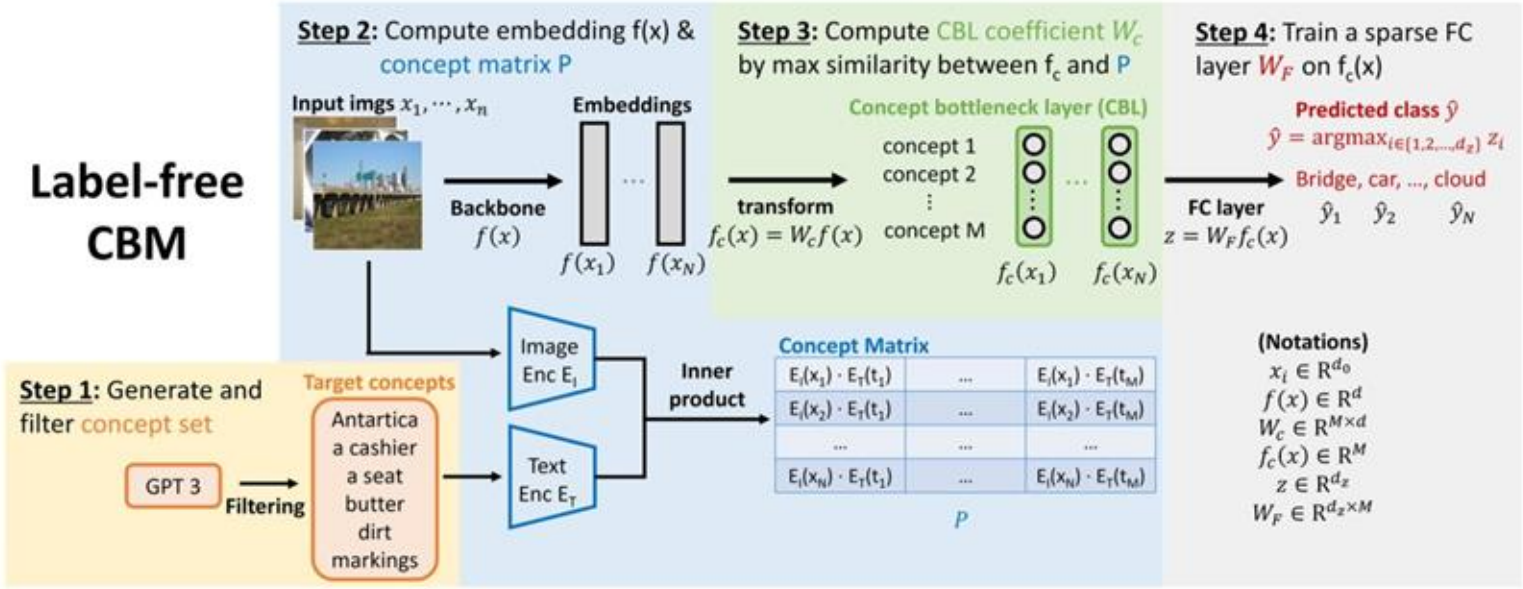
bird species

Prediction still flows through human-readable concepts, even when those concepts came from pseudo supervision.

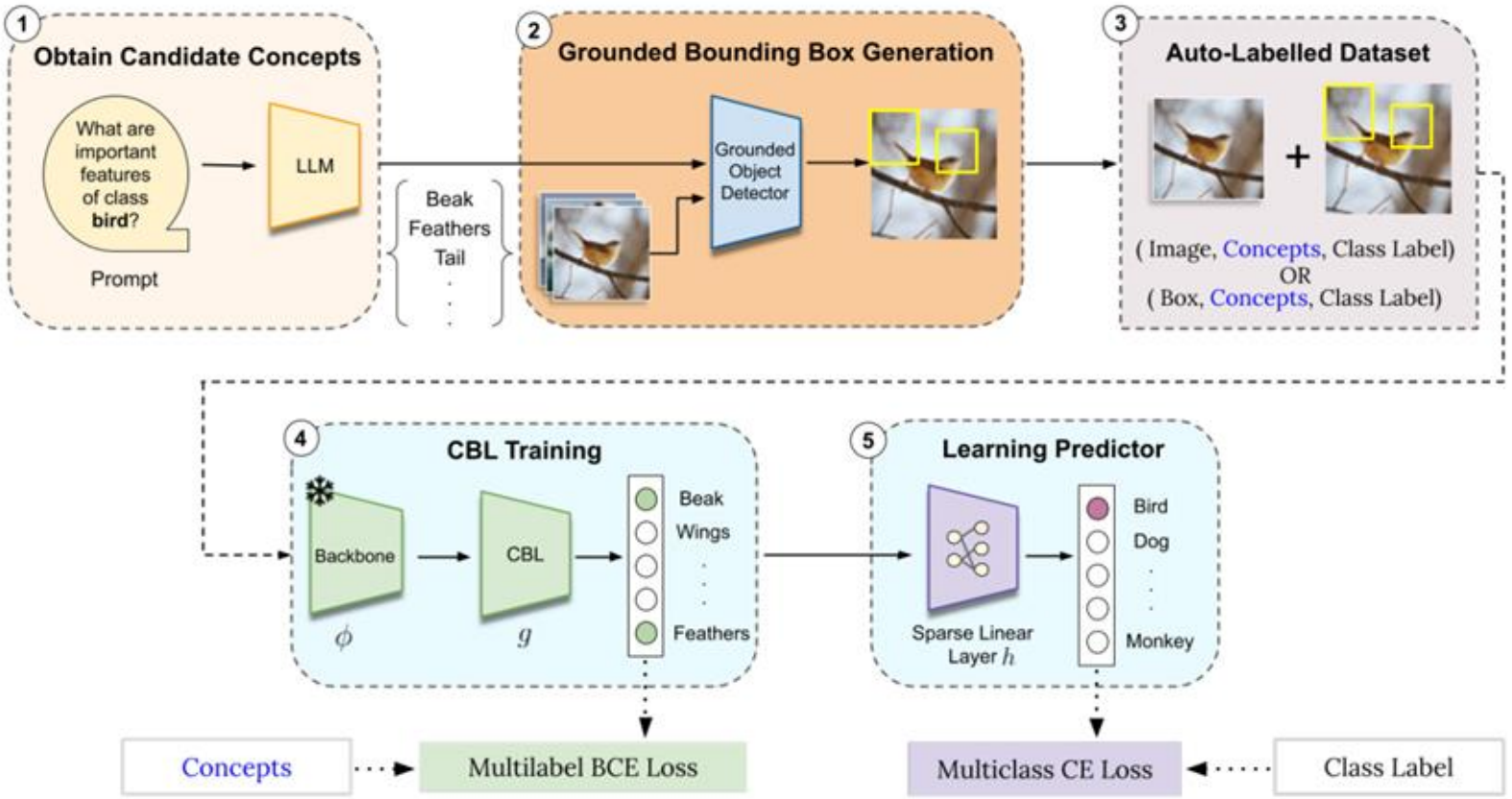
Caution: pseudo concepts reduce annotation cost, but they can be noisy or spurious

LF-CBM (ICLR 23)

- Learn the projection from feature space to concepts space
- Activation pattern for neuron k $q_k = [f_{c,k}(x_1), \ldots, f_{c,k}(x_N)]^\top$
- Cos-cubed loss: $$L(W_c) = \sum_{i=1}^{M} -\text{sim}(t_i, q_i) := \sum_{i=1}^{M} -\frac{\bar{q}_i^{\,3} \cdot \bar{P}_{:,i}^{\,3}}{||\bar{q}_i^{\,3}||_2 ||\bar{P}_{:,i}^{\,3}||_2}$$



VLG-CBM (NeurIPS 24)



Discover-then-name CBM (ECCV 24)

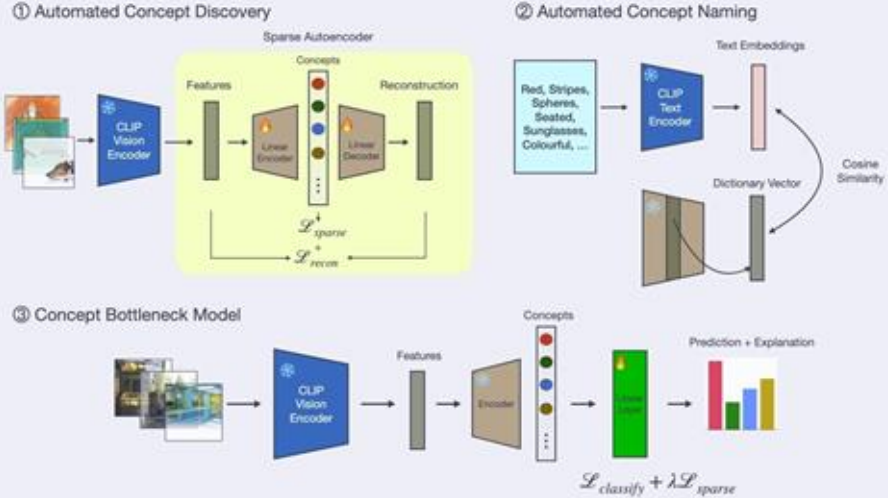


Fig. 2: Overview. Our approach consists of three steps. **(1)** We train a sparse autoencoder to extract disentangled concepts from a CLIP vision backbone. The autencoder is trained on a large dataset $\mathcal{D}_{extract}$ to reconstruct CLIP features using a linear combination of encoded concepts, which are optimized to be sparse using L_1 sparsity. The weights of the decoder can be interpreted as dictionary vectors whose linear sum with concept strengths reconstructs the original feature (Sec. 3.1). **(2)** We use a large concept set of texts $\mathcal{V}$ to name each extracted concept, by finding the text from the set whose embedding has the highest cosine similarity to concept's dictionary vector (Sec. 3.2). **(3)** We use the extracted and named concepts as a concept bottleneck layer, and train linear classifiers to construct inherently interpretable concept bottleneck models across downstream datasets $\mathcal{D}_{classify}$ using the same bottleneck layer (Sec. 3.3).

DCBM (ICML 25)

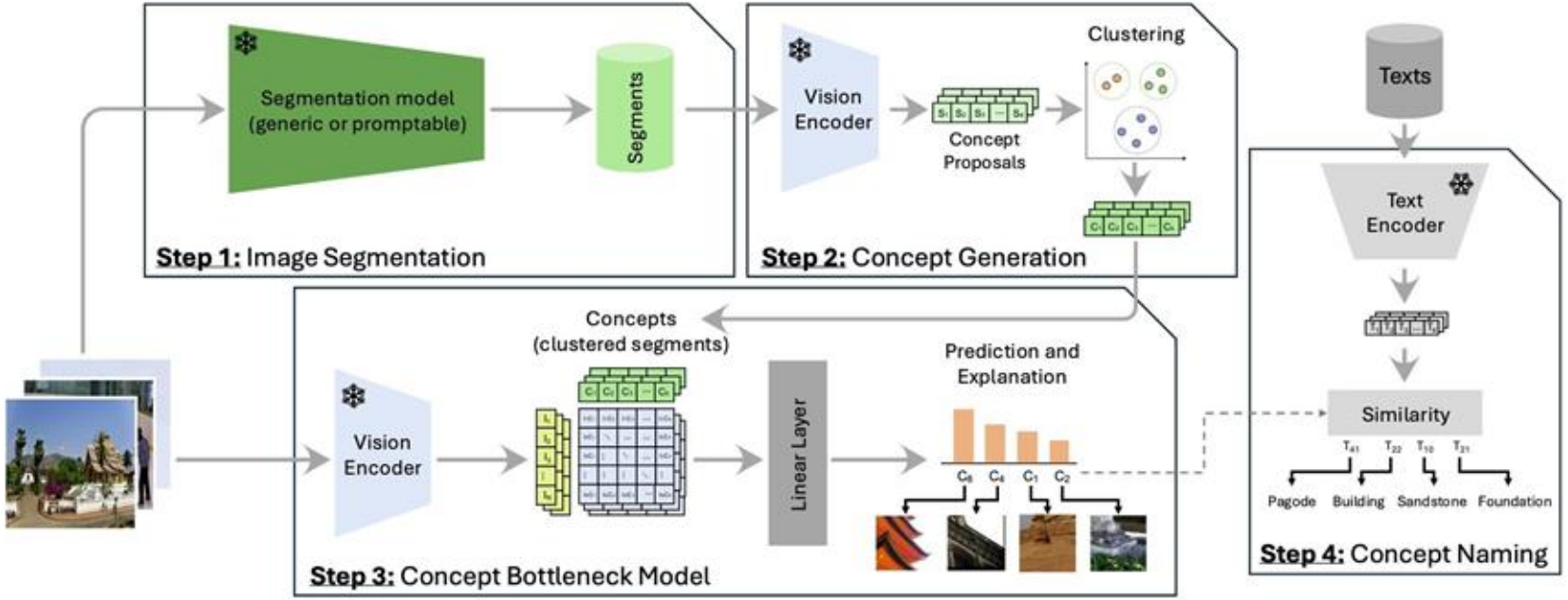
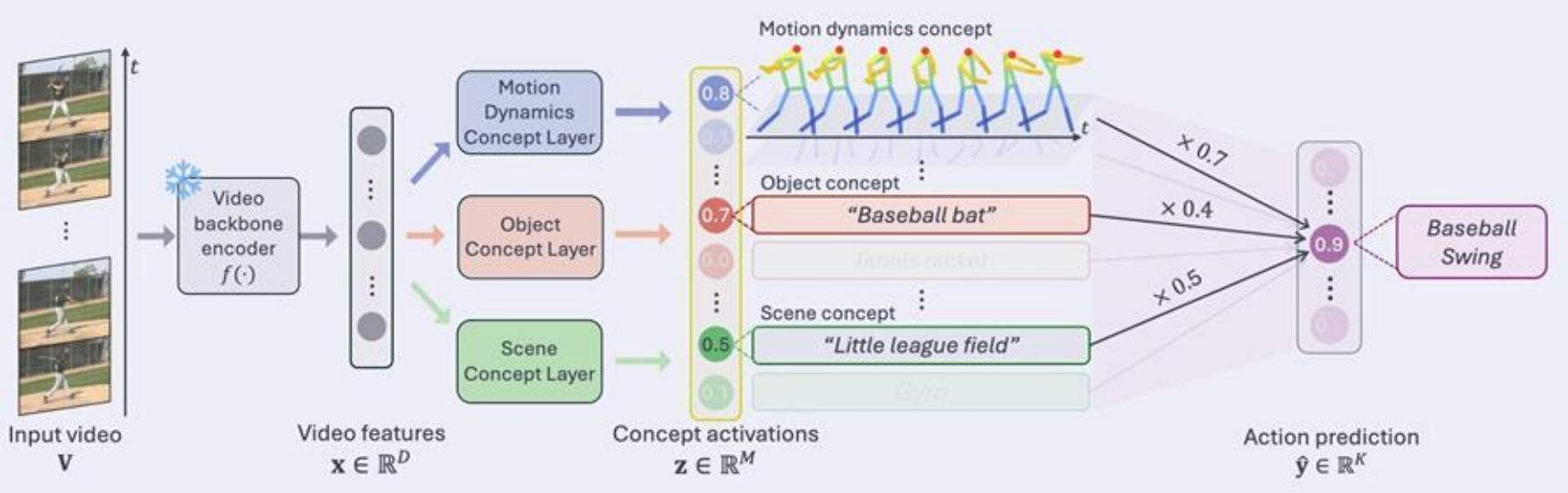


Figure 2. **DCBM framework.** The DCBM framework generates concept proposals through foundation models (Step 1). These proposals are then clustered, each represented by its centroid (Step 2). Finally, the unique concepts are utilized to train a sparse CBM, effectively (Step 3). We leverage the image-text alignment to map the visual concept to the corresponding textual concept (Step 4). We can remove undesired concepts after Step 2.

Participation-Aware Concept Bottleneck Models for Procedural Video

From object presence to action–object participation

DANCE (NeurIPS 2025)



three disentangled concept layers project the video features onto their own concept space-motion dynamics, object, and scene-producing disentangled activations

Task: procedural video classification

Example input video



A natural kitchen scene contains actions, manipulated objects, and unrelated context.

Input

A long, untrimmed RGB video

Training supervision

Coarse event timeline

Example annotation and target

SIL	take_cup	pour_coffee	pour_sugar	pour_milk	SIL

time →

Output

Coffee

The missing distinction in procedural explanations

Object presence is not the same as participation in an observed action.



Illustrative interpretation

POUR occurs

Cup / carton participate

Pan may be visible but unused

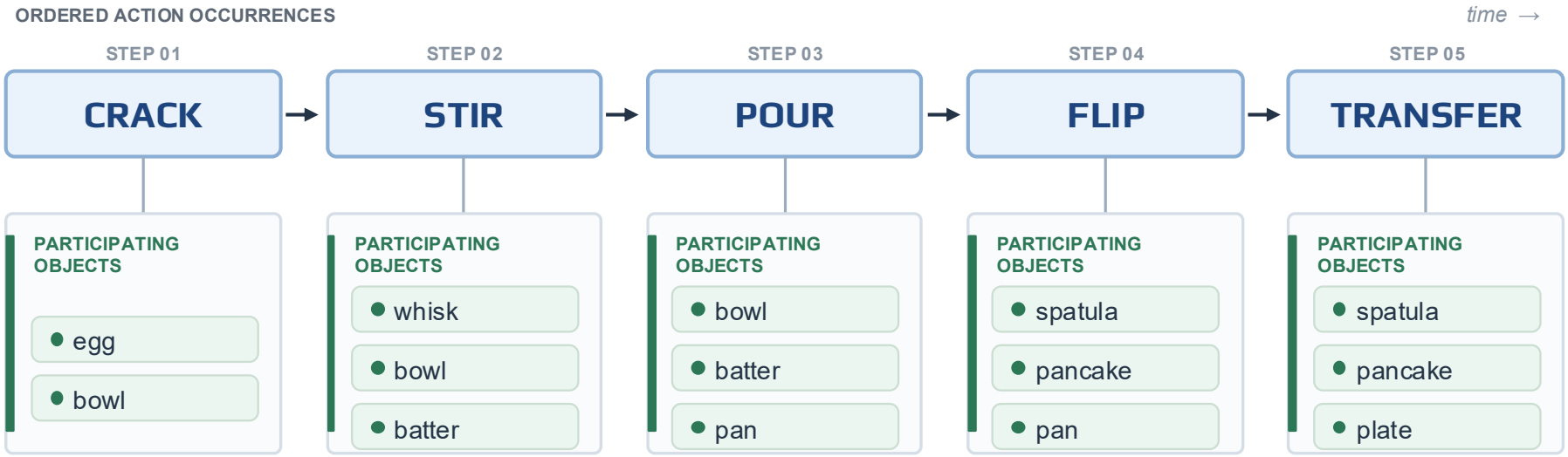
Research question

Given a recognized action occurrence, which visible objects actually participate in that occurrence?

Proposed Concept Representation

A procedure is an ordered sequence of actions, each linked to its participating objects.

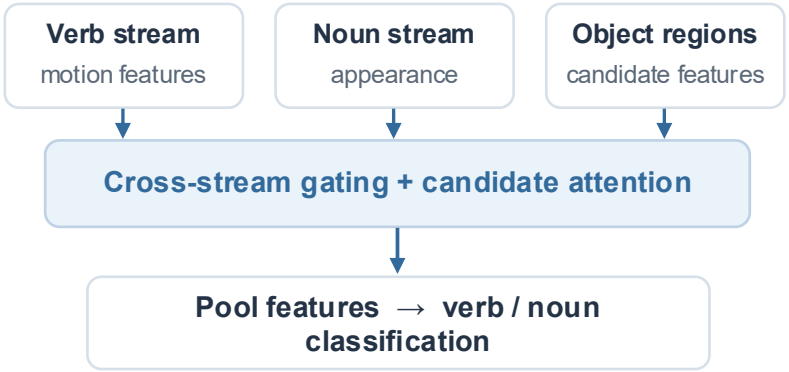
Example: pancakes



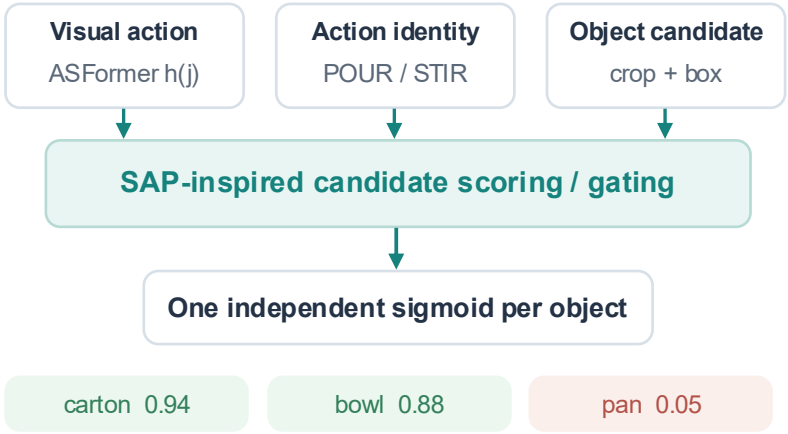
Visible does not mean participating.

From SAP / SAOA to participation prediction

SAP / SAOA: recognize a verb and noun

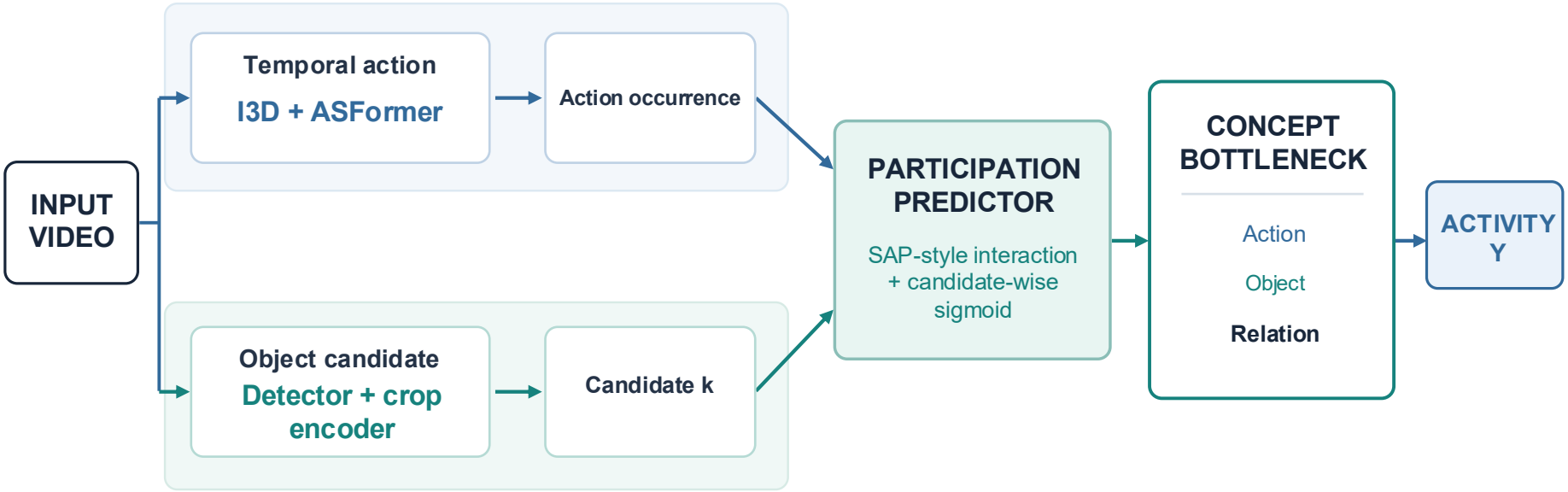


Ours: explain who participated in this action

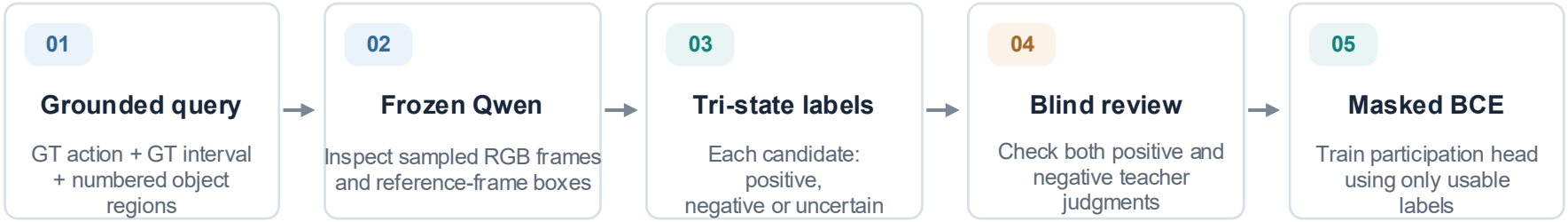


Sources: Wang et al., AAAI 2020 (SAP); Wang et al., IEEE TPAMI 2023 (SAOA).

Proposed architecture: participation-aware video CBM



Supervision: teach participation, then audit it



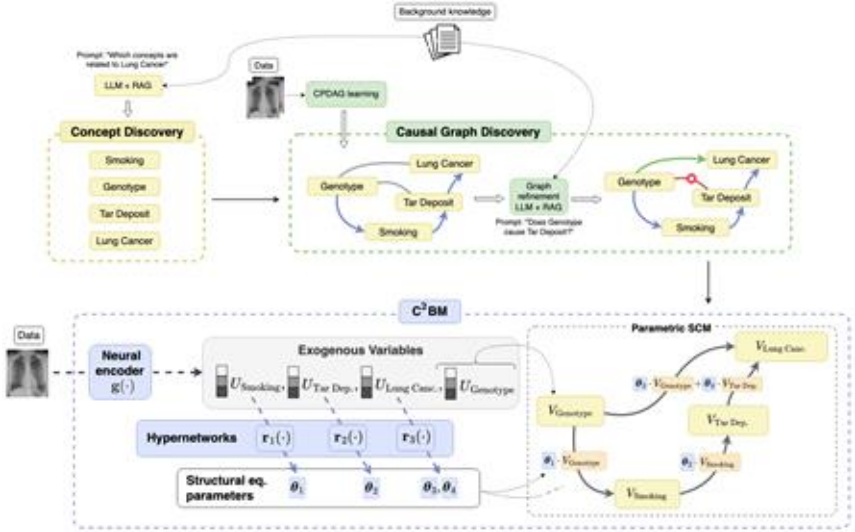
ANNOTATION TARGET (illustrative)

Milk carton	Participates
Receiving bowl	Participates
Unused pan	Does not participate
Occluded vessel	Uncertain

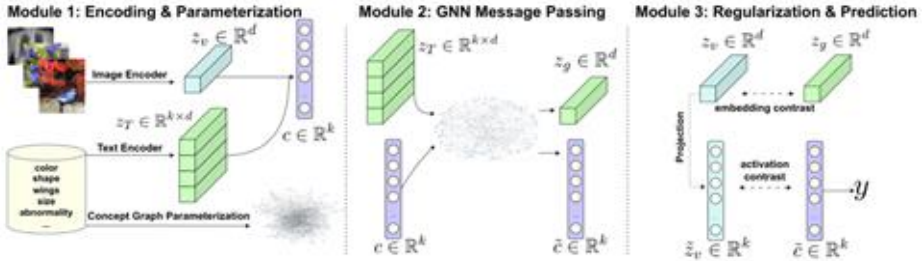
Research directions for CBM

1. Structured concept

Causally Reliable Concept Bottleneck Models(NeurIPS 2025)

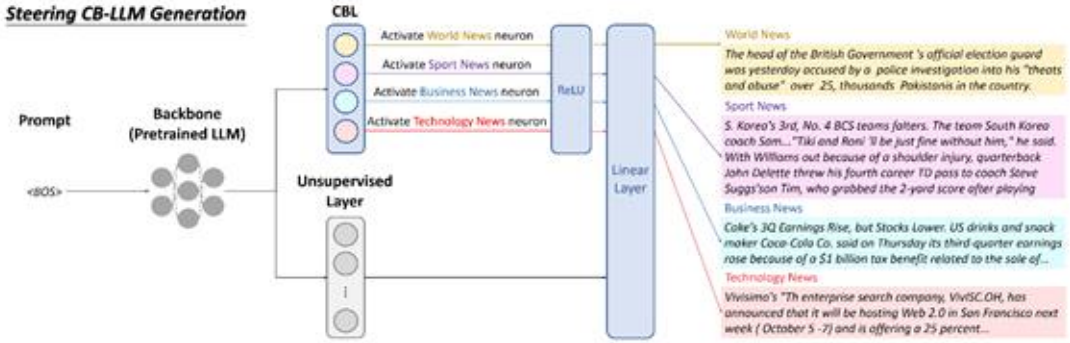


Graph Concept Bottleneck Models(TMLR2026)

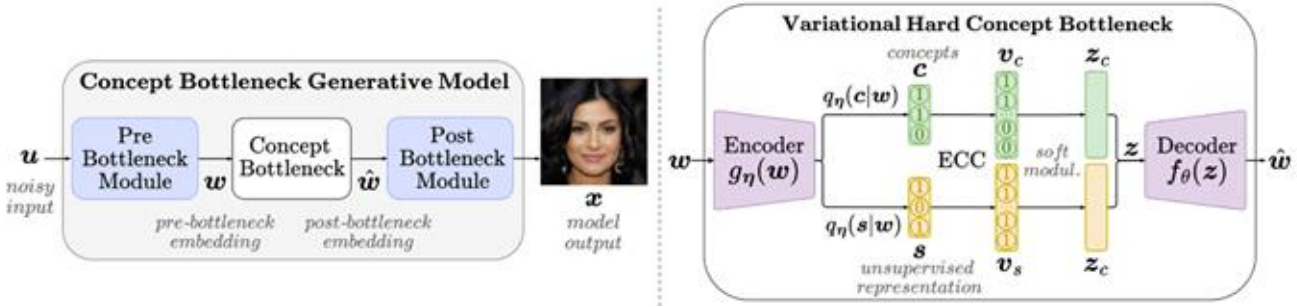


2. Applying bottleneck to other models

Concept Bottleneck Large Language Models(ICLR 2025)



A Probabilistic Hard Concept Bottleneck for Steerable Generative Models (ICLR 2026)



3. Leakage



blue_wing score

0.95

Bird A



blue_wing score

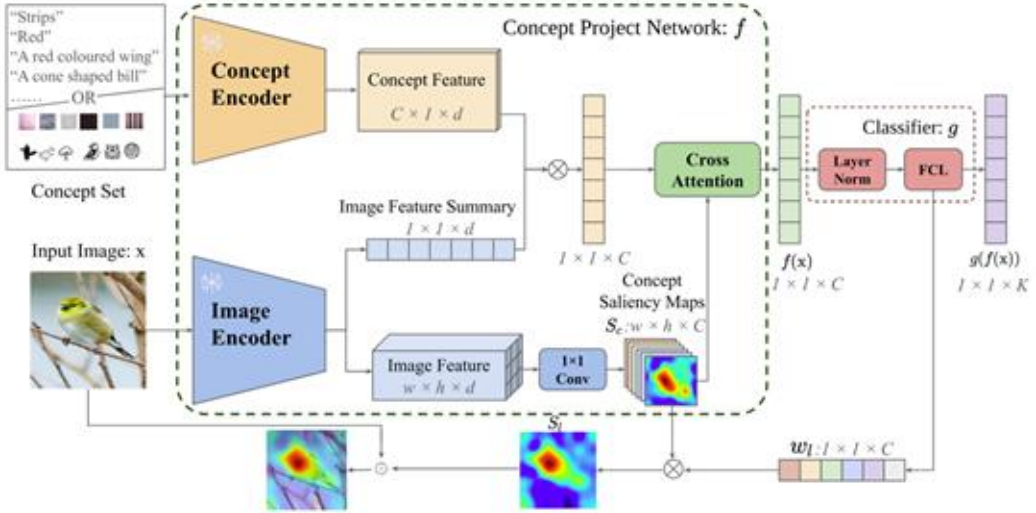
0.65

Bird B

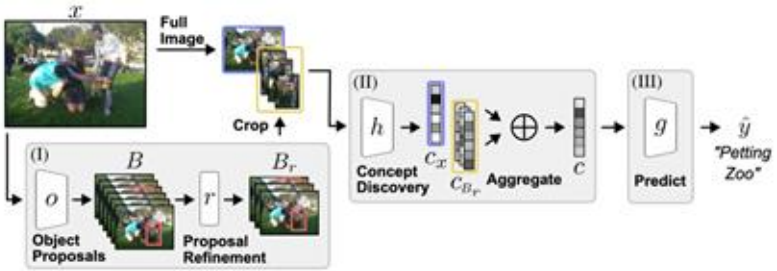
Problem: both birds may truly have blue wings, but the exact concept value can still encode extra hidden information that the classifier can exploit.

4. Locality

SL-CBM: Enhancing Concept Bottleneck Models with Semantic Locality(AAAI 2026)



Object-Centric Concept-Bottlenecks(NeurIPS 2025)



Thank you!